

KAIROUAN, Tunisia

WMO: 607350

Lat: 35.667N

Lon: 10.100E

Elev: 68

StdP: 100.51

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.0	6.1	-2.6	3.1	12.8	-1.1	3.5	12.7	7.7	14.6	6.4	14.3	1.1	220	0.350

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	41.9	22.0	39.8	21.8	37.9	21.6	25.2	32.4	24.5	31.8	23.8	31.4	2.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	18.5	27.8	22.6	17.5	27.6	21.8	16.6	27.2	77.3	32.5	74.2	31.8	71.5	31.4	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.2	5.3	4.5	DB	2.4	45.4	1.3	1.6	1.4	46.6	0.7	47.5	0.0	48.5	-0.9	49.6	(4)
			WB	0.9	27.0	1.1	1.3	0.1	27.9	-0.6	28.7	-1.2	29.5	-2.0	30.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	21.1	12.7	13.2	15.9	18.7	23.0	27.5	30.7	30.6	26.7	22.7	17.5	13.9	(6)
(7)		DBStd	6.96	2.29	2.57	3.05	2.73	2.96	3.10	2.32	2.46	2.89	2.83	2.59	2.36	(7)
(8)		HDD10.0	9	4	3	1	0	0	0	0	0	0	0	0	2	(8)
(9)		HDD18.3	624	176	145	88	27	1	0	0	0	2	46	139		(9)
(10)		CDD10.0	4078	88	93	183	262	403	526	643	638	501	394	224	122	(10)
(11)		CDD18.3	1649	1	1	12	39	146	276	384	380	251	138	19	1	(11)
(12)		CDH23.3	18772	6	19	147	354	1343	3201	5150	5005	2475	957	107	8	(12)
(13)	CDH26.7	10210	0	2	38	100	589	1821	3156	2983	1171	333	16	1	(13)	
(14)	Wind	WSAvg	1.8	1.6	1.9	1.9	2.1	2.1	2.0	1.7	1.6	1.4	1.5	1.6	(14)	
(15)	Precipitation	PrecAvg	292	24	22	32	26	22	11	3	13	44	43	26	25	(15)
(16)		PrecMax	735	121	108	95	67	89	54	32	101	191	428	103	127	(16)
(17)		PrecMin	134	0	0	2	0	0	0	0	0	2	0	0	0	(17)
(18)		PrecStd	107	25	22	24	17	21	12	5	18	41	60	25	29	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.0	26.0	30.9	32.2	38.8	44.1	44.6	44.2	39.8	35.2	28.9	24.2	(19)
MCWB			13.3	14.7	16.0	17.0	20.6	20.9	22.1	22.6	21.8	20.5	17.3	14.2	(20)	
2%		DB	21.1	22.9	26.9	29.4	34.9	39.7	41.8	41.8	36.8	32.2	26.1	21.9	(21)	
		MCWB	12.6	13.5	15.1	16.7	19.2	21.0	22.0	22.5	22.3	20.3	16.9	13.7	(22)	
5%		DB	19.0	20.6	24.5	27.1	32.1	37.0	39.8	39.4	34.8	30.1	24.1	20.1	(23)	
		MCWB	11.8	12.5	14.4	15.9	18.7	20.5	21.7	22.3	22.4	20.2	16.2	13.0	(24)	
10%		DB	17.3	18.8	22.1	25.1	29.9	34.9	37.8	37.4	32.7	28.1	22.6	18.8	(25)	
	MCWB	11.3	11.8	13.6	15.5	18.0	20.4	21.5	22.2	22.1	19.8	15.5	12.6	(26)		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.6	15.6	17.7	19.1	22.1	24.1	25.4	26.8	26.0	23.7	20.0	16.2	(27)
MCDWB			19.5	23.1	25.4	27.2	33.4	33.5	33.7	35.6	32.5	28.6	25.4	20.8	(28)	
2%		WB	13.6	14.2	16.3	17.7	20.6	22.8	24.5	25.5	24.9	22.6	18.6	14.9	(29)	
		MCDWB	18.5	20.6	23.8	25.2	30.7	32.6	32.7	33.1	30.6	27.0	22.7	18.9	(30)	
5%		WB	12.8	13.2	15.3	17.0	19.7	22.0	23.8	24.7	24.1	21.7	17.5	14.0	(31)	
		MCDWB	17.3	18.9	22.0	24.4	28.8	31.3	32.5	32.3	30.1	26.4	21.8	18.7	(32)	
10%		WB	12.0	12.4	14.4	16.2	18.9	21.3	23.1	24.0	23.2	20.9	16.5	13.2	(33)	
	MCDWB	16.6	17.6	20.4	22.8	27.3	31.0	32.4	31.8	29.5	26.0	21.1	17.5	(34)		
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.6	10.1	11.2	11.5	12.8	13.8	14.2	13.4	11.0	10.2	9.8	9.3	(35)
MCDBR			12.4	13.8	15.3	15.4	16.5	16.9	16.8	16.2	13.7	12.9	12.1	11.7	(36)	
MCWBR			6.4	6.7	6.2	5.7	5.1	4.6	4.6	4.6	4.2	4.6	5.5	6.0	(37)	
5% WB		MCDBR	10.5	12.1	13.3	13.2	14.5	14.9	14.6	13.8	12.0	10.8	10.1	9.9	(38)	
		MCWBR	5.9	6.2	5.9	5.2	5.0	4.9	4.7	4.5	4.5	4.3	5.3	5.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.337	0.364	0.431	0.481	0.503	0.473	0.498	0.489	0.483	0.439	0.384	0.347	(40)	
taud		2.379	2.292	2.084	1.973	1.943	2.056	2.001	2.024	2.050	2.130	2.254	2.358	(41)		
Ebn at Noon		852	868	832	809	795	818	794	793	773	771	783	813	(42)		
Edh at Noon		93	113	152	179	188	167	176	168	156	132	105	90	(43)		
(44)	All-Sky Solar Radiation	RadAvg	2.76	3.56	4.76	5.84	6.86	7.64	7.76	6.86	5.24	4.04	2.98	2.45	(44)	
(45)		RadStd	0.18	0.22	0.27	0.36	0.34	0.32	0.15	0.26	0.23	0.21	0.13	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.39	N/A	N/A	N/A	N/A	N/A	N/A	-73	+142	N/A
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page